

Daily Number Theory Drill #5 — Answers & Investigations

Sheet by Antigravity

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Bounding Diophantine Equations · Inequality Logic · Factor Searches.

Section A Rapid Recognition

(≤15 s each)

Last digits, primes, and basic properties. Pure recognition.

1. Find the last digit of 3^{2025} .

Answer: 3

Method: The last digits of powers of 3 follow a cycle of four: 3, 9, 7, 1. Since $2025 = 4 \times 506 + 1$, we have $2025 \equiv 1 \pmod{4}$. The last digit is the first in the cycle, which is 3.

Investigate further: Can you determine the last digit of 7^{2025} ? Hint: Find the cycle length for powers of 7.

2. Factorise 1001 completely into primes.

Answer: $7 \times 11 \times 13$

Method: Notice that 1001 is divisible by 11 ($1 - 0 + 0 - 1 = 0$). Dividing by 11 gives 91, and $91 = 7 \times 13$. Thus, the prime factorisation is $7 \times 11 \times 13$.

Investigate further: What is the prime factorisation of 1001001?

3. Find $\gcd(42, 105)$.

Answer: 21

Method: Factorise both numbers: $42 = 2 \times 3 \times 7$ and $105 = 3 \times 5 \times 7$. The common prime factors are 3 and 7, so $\gcd(42, 105) = 3 \times 7 = 21$.

Investigate further: What is the lowest common multiple of 42 and 105?

4. Find the number of positive divisors of 120.

Answer: 16

Method: Find the prime factorisation of 120: $120 = 2^3 \times 3^1 \times 5^1$. The number of divisors is $(3+1)(1+1)(1+1) = 4 \times 2 \times 2 = 16$.

Investigate further: What is the sum of the positive divisors of 120?

5. Evaluate $5^{10} \pmod{6}$.

Answer: 1

Method: Note that $5 \equiv -1 \pmod{6}$. Thus, $5^{10} \equiv (-1)^{10} = 1 \pmod{6}$.

Investigate further: What is $5^{11} \pmod{6}$?

6. Find the smallest positive integer x such that $2x \equiv 1 \pmod{5}$.

Answer: 3

Method: Multiply both sides by 3 (the inverse of 2 mod 5). This gives $6x \equiv 3 \pmod{5}$, and since $6 \equiv 1 \pmod{5}$, we get $x \equiv 3 \pmod{5}$. The smallest positive integer is 3.

Investigate further: What is the inverse of 3 modulo 7?

7. Find the smallest integer $n > 1$ such that $n \equiv 1 \pmod{3}$ and $n \equiv 1 \pmod{4}$.

Answer: 13

Method: By the Chinese Remainder Theorem, since $\gcd(3, 4) = 1$, the simultaneous congruences $n \equiv 1 \pmod{3}$ and $n \equiv 1 \pmod{4}$ imply $n \equiv 1 \pmod{12}$. The smallest integer $n > 1$ is $1 + 12 = 13$.

Investigate further: Find the smallest integer $n > 2$ such that $n \equiv 2 \pmod{3}$ and $n \equiv 2 \pmod{4}$.

8. Write the prime factorisation of 360.

Answer: $2^3 \times 3^2 \times 5$

Method: Break down 360 into $36 \times 10 = (2^2 \times 3^2) \times (2 \times 5) = 2^3 \times 3^2 \times 5$.

Investigate further: What is the prime factorisation of 720?

9. Calculate $7^3 \pmod{10}$.

Answer: 3

Method: Work modulo 10: $7^2 = 49 \equiv 9 \pmod{10}$, so $7^3 \equiv 9 \times 7 = 63 \equiv 3 \pmod{10}$.

Investigate further: What is the last digit of 7^4 ?

10. Find the largest prime factor of 9999.

Answer: 101

Method: Factorise 9999 as 9×1111 . Then $1111 = 11 \times 101$. Since 101 is prime, the prime factorisation is $3^2 \times 11 \times 101$, and the largest prime factor is 101.

Investigate further: What is the largest prime factor of 999999?

Section B Manipulation Drills

(≤ 50 s each)

Modular arithmetic, exponentiation, divisibility rules. Look for patterns.

1. The greatest power of 7 which is a factor of $50!$ is 7^k . What is k ? (after SMC 2023 Q12)

Answer: 8

Method: By Legendre's formula, the highest power of 7 dividing $50!$ is $\lfloor \frac{50}{7} \rfloor + \lfloor \frac{50}{49} \rfloor = 7 + 1 = 8$.

Investigate further: This question scales an SMC concept. How would you calculate the number of trailing zeroes of $50!$?

2. Find the smallest positive integer n such that $15n$ is a perfect cube.

Answer: 225

Method: We have $15n = 3^1 \times 5^1 \times n$. To make this a perfect cube, the exponents of all prime factors must be multiples of 3. We need at least 3^2 and 5^2 , so $n = 3^2 \times 5^2 = 9 \times 25 = 225$.

Investigate further: What is the smallest n such that $15n$ is a perfect square?

3. Find the remainder when $1! + 2! + 3! + \cdots + 100!$ is divided by 12.

Answer: 9

Method: Observe that for $k \geq 4$, $k!$ is a multiple of $4 \times 3 = 12$, so $k! \equiv 0 \pmod{12}$. The sum reduces to $1! + 2! + 3! = 1 + 2 + 6 = 9 \pmod{12}$.

Investigate further: Find the remainder when the sum is divided by 24.

4. Which is larger, 3^{40} or 4^{30} ?

Answer: 3^{40}

Method: Express both with the same exponent: $3^{40} = (3^4)^{10} = 81^{10}$ and $4^{30} = (4^3)^{10} = 64^{10}$. Since $81 > 64$, it follows that $3^{40} > 4^{30}$.

Investigate further: Which is larger: 2^{50} or 3^{30} ?

5. Find the smallest positive integer x such that $3x \equiv 4 \pmod{7}$.

Answer: 6

Method: Multiply both sides by 5 (the inverse of 3 mod 7, since $15 \equiv 1 \pmod{7}$). This gives $15x \equiv 20 \pmod{7}$, which simplifies to $x \equiv 6 \pmod{7}$.

Investigate further: Solve $4x \equiv 5 \pmod{7}$.

6. Find the last two digits of 7^{2022} .

Answer: 49

Method: By Euler's Totient Theorem, $\phi(100) = 40$, but we also know $7^4 = 2401 \equiv 1 \pmod{100}$. Since $2022 = 4 \times 505 + 2$, we have $7^{2022} \equiv 7^2 = 49 \pmod{100}$.

Investigate further: What are the last two digits of 3^{2022} ?

7. How many positive factors of 1000 are perfect squares?

Answer: 4

Method: The prime factorisation of 1000 is $2^3 \times 5^3$. A perfect square factor must be of the form $2^{2a} \times 5^{2b}$. The possible values for $2a$ are 0 and 2, and for $2b$ are 0 and 2. There are $2 \times 2 = 4$ such factors.

Investigate further: How many positive factors of 1000 are perfect cubes?

8. Find all integers x such that $0 \leq x < 8$ and $x^2 \equiv 1 \pmod{8}$.

Answer: 1, 3, 5, 7

Method: Test the odd numbers mod 8: $1^2 = 1 \equiv 1$, $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, $7^2 = 49 \equiv 1$. All four odd residues satisfy the congruence.

Investigate further: Solve $x^2 \equiv 1 \pmod{16}$.

9. Given $2^x 3^y = 12^3$, find $x + y$.

Answer: 9

Method: Expand the right hand side: $12^3 = (2^2 \times 3)^3 = 2^6 \times 3^3$. By comparing exponents, $x = 6$ and $y = 3$. Hence $x + y = 9$.

Investigate further: If $2^x 5^y = 100^2$, what is $x + y$?

10. Find all pairs of positive integers (x, y) such that $xy = x + y + 3$.

Answer: (2, 5), (3, 3), (5, 2)

Method: Rearrange to $xy - x - y = 3$. Add 1 to both sides to complete the rectangle: $xy - x - y + 1 = 4$, which factors as $(x - 1)(y - 1) = 4$. Since $x, y \geq 1$, the pairs for $(x - 1, y - 1)$ are (1, 4), (2, 2), (4, 1). Thus (x, y) can be (2, 5), (3, 3), (5, 2).

Investigate further: Find the integer solutions to $xy = x + y + 5$.

Section C Substitution & Structure

(≤90 s each)

Synoptic links: use algebraic identities and combinatorial counting to solve these number theory problems.

1. The expression $\frac{n^2+3n+5}{n+1}$ takes integer values for certain positive integer values of n . What is the sum of all such positive integer values of n ? (after SMC 2023 Q19)

Answer: 2

Method: Perform polynomial division: $\frac{n^2+3n+5}{n+1} = \frac{n(n+1)+2(n+1)+3}{n+1} = n + 2 + \frac{3}{n+1}$. For this to be an integer, $n + 1$ must divide 3. The positive divisors of 3 are 1 and 3, yielding $n = 0$ and $n = 2$. The only positive integer is $n = 2$, so the sum is 2.

Investigate further: This adapts a core SMC structural problem. How would the answer change if n could be any integer?

2. How many of the numbers 6, 7, 8, 9, 10 are factors of the sum $2^{2024} + 2^{2023} + 2^{2022}$? (after SMC 2023 Q10)

Answer: 2

Method: Factor out the lowest power: $2^{2022}(2^2 + 2^1 + 1) = 7 \times 2^{2022}$. Examining the options: 7 divides it, 8 divides it (since 2^3 divides 2^{2022}), but 6, 9, and 10 do not since they require a factor of 3 or 5.

Investigate further: This questions mirrors SMC factorisation themes. Is 14 a factor?

3. Find all pairs of positive integers (x, y) such that $x^2 - y^2 = 17$.

Answer: (9, 8)

Method: Factorise the difference of squares: $(x-y)(x+y) = 17$. Since 17 is prime and $x+y > x-y > 0$, we must have $x-y = 1$ and $x+y = 17$. Adding gives $2x = 18 \implies x = 9$, and $y = 8$.

Investigate further: How many solutions exist if 17 is replaced by 15?

4. How many ordered pairs of positive integers (a, b) satisfy $\text{lcm}(a, b) = 300$?

Answer: 75

Method: Prime factorise 300: $300 = 2^2 \times 3^1 \times 5^2$. Let $a = 2^{x_1} 3^{y_1} 5^{z_1}$ and $b = 2^{x_2} 3^{y_2} 5^{z_2}$. We need $\max(x_1, x_2) = 2$, $\max(y_1, y_2) = 1$, and $\max(z_1, z_2) = 2$. The number of pairs for an exponent k is $2k + 1$. So $(2(2) + 1) \times (2(1) + 1) \times (2(2) + 1) = 5 \times 3 \times 5 = 75$.

Investigate further: How many ordered pairs satisfy $\text{lcm}(a, b) = 1000$?

5. How many positive integers $n \leq 100$ have exactly 3 positive divisors?

Answer: 4

Method: A number has exactly 3 positive divisors if and only if it is the square of a prime. The primes whose squares are ≤ 100 are 2, 3, 5, 7. Thus, there are 4 such numbers.

Investigate further: How many numbers ≤ 100 have exactly 4 positive divisors?

6. Find the sum of the digits of the largest integer n such that $n^3 + 100$ is a multiple of $n + 10$.

Answer: 17

Method: Use polynomial division: $n^3 + 100 = (n^3 + 1000) - 900 = (n + 10)(n^2 - 10n + 100) - 900$. Thus, $n + 10$ must divide -900 . The largest integer n corresponds to the largest positive factor of 900, which is 900. Hence $n + 10 = 900 \implies n = 890$. The sum of the digits is $8 + 9 + 0 = 17$.

Investigate further: Find the smallest integer n satisfying this condition.

7. Find all triples of positive integers (x, y, z) such that $x \leq y \leq z$ and $xyz = x + y + z$.

Answer: (1, 2, 3)

Method: Since $x \leq y \leq z$, we have $x + y + z \leq 3z$. Therefore $xyz \leq 3z \implies xy \leq 3$. Since x and y are positive integers with $x \leq y$, the possible pairs for (x, y) are (1, 1), (1, 2), and (1, 3). If (1, 1), $z = 2 + z$ (impossible). If (1, 2), $2z = 3 + z \implies z = 3$. If (1, 3), $3z = 4 + z \implies z = 2$, but we need $y \leq z$. The only solution is (1, 2, 3).

Investigate further: This builds upon BMO1 Diophantine constraints. Can you find all positive integer solutions (x, y, z, w) to $xyzw = x + y + z + w$ with $x \leq y \leq z \leq w$?

8. Find the sum of all integer values of n for which $n^2 + 4n + 3$ is a perfect square. (after AI Prompts D5)

Answer: -4

Method: Set $n^2 + 4n + 3 = k^2$. Complete the square: $(n + 2)^2 - 1 = k^2 \implies (n + 2)^2 - k^2 = 1$. The difference of two squares is 1 only when they are 1 and 0, so $(n + 2)^2 = 1 \implies n + 2 = \pm 1 \implies n = -1, -3$. The sum is $(-1) + (-3) = -4$.

Investigate further: This expands an algebraic pattern into Diophantine analysis. Does $n^2 + 6n + 5$ have similar properties?

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find all prime numbers p such that $p^2 + 2$ is also a prime number.

Answer: 3

Method: If $p = 3$, $3^2 + 2 = 11$, which is prime. For $p > 3$, $p \equiv 1$ or $2 \pmod{3}$, which means $p^2 \equiv 1 \pmod{3}$. Then $p^2 + 2 \equiv 3 \equiv 0 \pmod{3}$. Since $p^2 + 2 > 3$, it must be a composite multiple of 3. So $p = 3$ is the only solution.

Investigate further: If $p^2 + 2$ is changed to $p^2 + 4$, are there any prime solutions?

2. Find all pairs of positive integers (x, y) such that $x^2y = x^2 + 3y + 2$.

Answer: (2, 6)

Method: Rearrange to isolate y : $y(x^2 - 3) = x^2 + 2 \implies y = \frac{x^2+2}{x^2-3} = 1 + \frac{5}{x^2-3}$. For y to be an integer, $x^2 - 3$ must be a factor of 5. The factors of 5 are $\pm 1, \pm 5$. Setting $x^2 - 3$ to 1, $-1, 5, -5$ yields $x^2 = 4, 2, 8, -2$. The only perfect square is $x^2 = 4$, so $x = 2$ (since $x > 0$). This gives $y = 1 + 5/1 = 6$. The only pair is (2, 6).

Investigate further: This explores algebraic rearrangement followed by a factor search. Find all integer pairs (x, y) satisfying $x^2y = x^2 + 4y + 3$.

3. How many pairs of positive integers (x, y) with $x \leq y$ satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$?

Answer: 5

Method: Since $x \leq y$, $\frac{1}{y} \leq \frac{1}{x}$. Thus $\frac{1}{6} = \frac{1}{x} + \frac{1}{y} \leq \frac{2}{x}$, which implies $x \leq 12$. We also need $\frac{1}{x} < \frac{1}{6}$, so $x > 6$. Testing $x \in \{7, 8, 9, 10, 11, 12\}$ yields integer y for $x = 7, 8, 9, 10, 12$. This gives 5 pairs: (7, 42), (8, 24), (9, 18), (10, 15), (12, 12).

Investigate further: This question tests bounding Diophantine equations instead of using Simon's Favourite Factoring Trick. How many pairs (x, y) with $x \leq y$ satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{p}$ for a prime p ?

4. Find the remainder when $1^{2025} + 2^{2025} + \dots + 10^{2025}$ is divided by 11.

Answer: 0

Method: By Fermat's Little Theorem, $x^{10} \equiv 1 \pmod{11}$ for $1 \leq x \leq 10$. Thus $x^{2025} = (x^{10})^{202} \times x^5 \equiv x^5 \pmod{11}$. Modulo 11, we have $(-x)^5 = -x^5 \equiv 11 - x^5$, so the terms pair up and cancel: $1^5 + 10^5 \equiv 1^5 + (-1)^5 = 0 \pmod{11}$. This is true for all pairs, leaving a remainder of 0.

Investigate further: Evaluate $1^{2026} + 2^{2026} + \dots + 10^{2026} \pmod{11}$.

5. Find the sum of all integers n for which $n^4 + 4$ is a prime number.

Answer: 0

Method: Apply the Sophie Germain Identity by adding and subtracting $4n^2$: $n^4 + 4 = (n^2 + 2)^2 - (2n)^2 = (n^2 - 2n + 2)(n^2 + 2n + 2)$. For this product to be prime, the smaller factor must be 1 or -1. Since

$n^2 \pm 2n + 2 = (n \pm 1)^2 + 1 \geq 1$, we set $(n - 1)^2 + 1 = 1 \implies n = 1$, yielding $1 \times 5 = 5$ (prime), and $(n + 1)^2 + 1 = 1 \implies n = -1$, yielding $5 \times 1 = 5$ (prime). The sum of n is $1 + (-1) = 0$.

└ **Investigate further:** Use the identity to factorise $a^4 + 4b^4$.

“Number theory is the queen of mathematics.” — Carl Friedrich Gauss

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths